

# HDOC Day-13

## Question: Menu-Driven Program for Even and Odd Digit Analysis

### 1. Problem Statement & Identification

- **Problem:** Create a menu-driven program using switch-case to input a positive integer and perform one of the following operations:
  1. Count the number of even digits and odd digits in the number.
  2. Calculate the sum of even digits and the sum of odd digits separately.
- **Input:** A positive integer (num) and menu choice (choice).
- **Output:** Results corresponding to the selected menu option.

### 2. Algorithm

1. **Start**
2. Input a positive integer num.
3. Display menu options (1 and 2).
4. Read user choice.

5. Execute operations based on choice using switch-case:

- **Case 1:**

- 1. Set temp = num, even\_dig = 0, odd\_dig = 0.

- 2. While temp > 0:

- Extract digit digit = temp % 10.
    - If digit % 2 == 0, increment even\_dig.
    - Else, increment odd\_dig.
    - Update temp = temp / 10.

- 3. Print even\_dig and odd\_dig.

- **Case 2:**

- 1. Set temp = num, sum\_even = 0, sum\_odd = 0.

- 2. While temp > 0:

- Extract digit digit = temp % 10.
    - If digit % 2 == 0, add digit to sum\_even.
    - Else, add digit to sum\_odd.
    - Update temp = temp / 10.

- 3. Print sum\_even and sum\_odd.

- **Default:** Print "Invalid Choice!".

6. **End**

### 3. Test Case Table

| <b>Test Case</b> | <b>Input Number</b> | <b>Choice</b> | <b>Operation</b> | <b>Expected Output</b>       | <b>Actual Output</b>         | <b>Status</b> |
|------------------|---------------------|---------------|------------------|------------------------------|------------------------------|---------------|
| <b>1</b>         | 12045               | 1             | Count Even/Odd   | Even_dig = 3,<br>Odd_dig = 2 | Even_dig = 3,<br>Odd_dig = 2 | Pass          |
| <b>2</b>         | 12345               | 2             | Sum Even/Odd     | Sum_even = 6,<br>Sum_odd = 9 | Sum_even = 6,<br>Sum_odd = 9 | Pass          |
| <b>3</b>         | 2468                | 1             | Count Even/Odd   | Even_dig = 4,<br>Odd_dig = 0 | Even_dig = 4,<br>Odd_dig = 0 | Pass          |

| <b>Tes<br/>t<br/>Ca<br/>se</b> | <b>Input<br/>Num<br/>ber</b> | <b>Cho<br/>ice</b> | <b>Opera<br/>tion</b> | <b>Expec<br/>ted<br/>Outpu<br/>t</b>        | <b>Actua<br/>l<br/>Outpu<br/>t</b>          | <b>Stat<br/>us</b> |
|--------------------------------|------------------------------|--------------------|-----------------------|---------------------------------------------|---------------------------------------------|--------------------|
| <b>4</b>                       | 1357                         | 2                  | Sum<br>Even/<br>Odd   | Sum_<br>even =<br>0,<br>Sum_<br>odd =<br>16 | Sum_<br>even =<br>0,<br>Sum_<br>odd =<br>16 | Pas<br>s           |

#### 4. C Source Code

```
#include <stdio.h>

int main(void) {
    int choice, num, temp, digit;
    int even_dig = 0, odd_dig = 0;
    int sum_even = 0, sum_odd = 0;

    printf("Enter a positive integer: ");
    if (scanf("%d", &num) != 1 || num <= 0) {
        printf("Invalid input! Please enter a positive integer.\n");
        return 1;
    }

    printf("\n--- MENU ---\n");
    printf("1. Count the number of even digits and odd digits\n");
    printf("2. Find the sum of even digits and sum of odd digits  
separately\n");
    printf("Enter your choice (1 or 2): ");
    scanf("%d", &choice);

    switch (choice) {
```

```

case 1:
    temp = num;
    while (temp > 0) {
        digit = temp % 10;
        if (digit % 2 == 0) {
            even_dig++;
        } else {
            odd_dig++;
        }
        temp /= 10;
    }
    printf("\nNumber: %d, Even_dig = %d, Odd_dig = %d\n", num,
even_dig, odd_dig);
    break;

case 2:
    temp = num;
    while (temp > 0) {
        digit = temp % 10;
        if (digit % 2 == 0) {
            sum_even += digit;
        } else {
            sum_odd += digit;
        }
        temp /= 10;
    }
    printf("\nNumber: %d, Sum_even = %d, Sum_odd = %d\n", num,
sum_even, sum_odd);
    break;

default:
    printf("\nInvalid Choice!\n");
    return 1;
}

return 0;
}

```

## 5. Compilation, Execution & Screenshots

```
kali@kali: ~/HDOOC_Day_13
Session Actions Edit View Help
kali@kali:~$ cd HDOOC_Day_13
kali@kali:~/HDOOC_Day_13$ nano q1.c
kali@kali:~/HDOOC_Day_13$ gcc q1.c -o q1
./q1
Enter a positive integer: 12045

--- MENU ---
1. Count the number of even digits and odd digits
2. Find the sum of even digits and sum of odd digits separately
Enter your choice (1 or 2): 1

Number: 12045, Even_dig = 3, Odd_dig = 2
kali@kali:~/HDOOC_Day_13$
```